MACHINE LEARNING PBL PROJECT

Online Game Price Prediction Using Machine Learning

Project Report and Implementation Plan

Course	Machine Learning (PBL)
Project Domain	Gaming / E-commerce
Learning Task	Supervised regression
Dataset	Steam Games - UOC Project (Zenodo)

PROJECT TEAM

Member	PRN
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Dataset snapshot: March 2025 | Currency: EUR

1. Project Overview

This project develops a beginner-friendly machine learning regression workflow that estimates the observed Steam game price from selected game characteristics. The system includes dataset cleaning, exploratory data analysis, feature preparation, model comparison, held-out evaluation, and a Streamlit interface.

2. Problem Statement

Online games are listed at different prices, and the prices seen on a storefront may depend on game characteristics and market conditions. This project investigates whether selected non-price fields in a public Steam dataset can help estimate the observed price recorded in that dataset. The prediction is an educational estimate, not a guarantee of a game's current or future price.

3. Objectives and Scope

- Download and document a relevant public labelled dataset.
- Inspect data quality and handle missing, invalid, duplicate, and inconsistently formatted values.
- Create explanatory EDA charts and interpret only patterns supported by the observed data.
- Extract launch year and encode categorical inputs without using target-related fields as predictors.
- Compare a simple baseline and multiple regression algorithms using cross-validation.
- Evaluate the selected model on a held-out test set with MAE, RMSE, and R^2.
- Deliver a Streamlit interface, executable notebook, source code, README, report, and presentation.

Scope boundary: the initial model estimates the `price` column in euros. It does not scrape Steam live, forecast future sales, predict game quality, or estimate prices for other regions.

4. Dataset and Data Dictionary

Source: Carlos Díez García and Pedro Vega Navarro, "Steam Games - UOC Project," Zenodo, published 1 April 2025. The dataset record describes 3,537 games collected in March 2025 from the Spanish-language Steam web store. The source lists the following fields. Dataset source: https://zenodo.org/records/15121027. DOI: https://doi.org/10.5281/zenodo.15121027. The Zenodo record specifies CC BY-SA 4.0; attribution and license terms should be preserved.

Column	Meaning	Use in this project
title	Game title	Excluded: identifier-like text
genres	Game genre string	Input feature
description	Store description	Excluded from baseline model
url	Store page URL	Excluded
price	Observed price in euros	Target variable
discount	Discount percentage	Excluded to reduce leakage
original_price	Price before discount	Excluded to reduce leakage
launch_date	Launch date	Used to derive launch_year
is_error	Scraping error flag	Rows flagged as errors are filtered where available
reviews	Number of reviews	Input feature
general_reviews	General review summary	Input feature
tags	Game tags	Excluded from initial baseline

5. Data Preprocessing and Cleaning

1. Read the source CSV with delimiter detection and normalise column names.
2. Parse numeric price values, including decimal-comma currency strings and free-game labels.
3. Remove rows flagged as scraping errors when the `is_error` column is present.
4. Check exact duplicate rows and remove duplicates.
5. Parse review counts, fill missing review counts with zero, and handle missing genre/review-summary values as “Unknown”.
6. Extract year from `launch_date`; invalid years are treated as missing and imputed inside the preprocessing pipeline.
7. Drop records whose target price is missing or invalid; retain free games as price zero.
8. Fit imputation, scaling, and one-hot encoding only inside training pipelines to prevent leakage.

6. Exploratory Data Analysis

The notebook generates these charts after the dataset is downloaded:

- Histogram of observed game price in EUR.
- Counts and median prices for the most frequent genre strings.
- Scatter plot of review count versus observed price.
- Missing-value and unique-value summary, plus duplicate and error-row counts.

The team must write interpretations from the actual plots after running the notebook. No chart trend or numeric finding is pre-claimed in this document.

7. Feature Engineering and Leakage Prevention

The first model uses four fields: `genres`, `launch_year`, `reviews`, and `general_reviews`. Year is derived from the launch date. Numeric fields are imputed and scaled; categorical fields are imputed and one-hot encoded. The `price` field is the target. `discount` and `original_price` are excluded from inputs because they directly relate to the price target. Titles, URLs, description text, and tags are also excluded from the initial baseline to keep the model interpretable and manageable.

8. Model Development

Model	Role	Selection approach
Median Baseline	Reference point	5-fold CV MAE
Ridge Regression	Linear baseline	5-fold CV MAE
Decision Tree Regressor	Nonlinear single tree	5-fold CV MAE
Random Forest Regressor	Ensemble of trees	5-fold CV MAE
Extra Trees Regressor	Randomised tree ensemble	5-fold CV MAE

Use an 80/20 train-test split with random seed 42. Compare models using five-fold cross-validation MAE on the training split. Select the candidate with the lowest mean CV MAE; fit it on the training split and use the held-out test set once for final evaluation.

9. Evaluation and Results

The training script writes the actual comparison and evaluation after execution. Enter the results from `reports/model_comparison.csv` and `reports/test_metrics.json` into the final submission. Do not invent scores.

Metric	Meaning	Actual result
MAE	Average absolute error in euros	Pending training run
RMSE	Penalises larger errors more strongly	Pending training run
R^2	Explained variance score	Pending training run
Best model	Lowest 5-fold CV MAE	Pending training run

10. Business Insights and Recommendations

After evaluation, discuss model performance in euros and describe which types of records show larger errors using the generated predictions. Any recommendation should be limited to dataset-supported patterns. The model cannot establish causal relationships or real-world commercial impact, and it should not be presented as a live price tool.

11. Deployment / Application

The Streamlit app accepts a genre string, launch year, review count, and review-summary category. It loads `models/best_model.joblib` and `models/model_metadata.json`, then displays the predicted observed price in euros and the final test metrics. Run `streamlit run app.py` from the repository root after downloading the dataset and training the model.

12. Implementation Plan

Phase	Tasks	Evidence/output
1. Setup	Create environment; install packages; download dataset	Working environment and raw CSV
2. Data understanding	Inspect columns, data types, missing values, duplicates	Data dictionary and quality summary
3. Preprocessing + EDA	Clean fields; create charts; document insights	Notebook outputs and figures
4. Model development	Train baseline and four regressors; cross-validate	Model comparison CSV
5. Evaluation	Select by CV MAE and test once on held-out split	Test metrics and actual-vs-predicted plot
6. Application	Run prediction app; test valid and invalid inputs	Streamlit demo
7. Submission	Finalise notebook, report, PPT, README and source	Complete submission package

13. Rubric Coverage (100 Marks)

Evaluation criterion	Marks
Problem Definition and Business Understanding	10
Dataset and Data Understanding	10
Data Preprocessing and Cleaning	10
Exploratory Data Analysis	10
Feature Engineering and Selection	10
Model Selection and Implementation	15
Model Evaluation and Validation	15
Business Insights and Recommendations	10
Deployment / Application	5
Documentation and Presentation	5
TOTAL	**100**

14. Deliverables Checklist

☐ Problem statement and objectives.

☐ Dataset source link and data dictionary.

☐ Complete executable Jupyter Notebook or Python source code.

☐ Preprocessing and EDA results with visualisations.

☐ Feature engineering and model implementation.

☐ Evaluation results with interpretation.

☐ Business insights and recommendations.

☐ Working application or deployment demonstration.

☐ Structured project report, presentation, organised source code, and README with execution instructions.

15. References

1. Díez García, C., & Vega Navarro, P. (2025). Steam Games - UOC Project [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.15121027

2. scikit-learn documentation: https://scikit-learn.org/stable/

3. Streamlit documentation: https://docs.streamlit.io/

Dataset snapshot and limitation: the source record describes a collection made in March 2025 from the Spanish-language Steam web store. Prices are stored in euros and may not match a user's current regional price.